

Project 1: AI for acne detection and classification

Content	
Project Title	AI for acne detection and classification
Category	Derma
Problem Statement	Dermatology is the branch of medicine concerned with diagnosis and treatment of skin disorders. AI has the potential to improve dermatologic care, especially in underserved communities. AI is valuable in analyzing skin images and patient information, enabling faster diagnosis and the more accurate identification of skin lesions. Challenges remain in implementation, safety, ethical issues and trustworthiness. Blackheads, Whiteheads, Papules, Pustules, and Cysts are the 5 major types of acne world-wide. The problem is to accurately detect and classify acne using images of patients.
Tech Stack	Gen AI, Python
Max Groups	4

Project 2: AI for cosmetic dermatology viz face beautification

Content	
Project Title	AI for cosmetic dermatology viz face beautification
Category	Derma
Problem Statement	Dermatology is the branch of medicine concerned with diagnosis and treatment of skin disorders. AI has the potential to improve dermatologic care, especially in underserved communities. AI is valuable in analyzing skin images and patient information, enabling faster diagnosis and the more accurate identification of skin lesions. Challenges remain in implementation, safety, ethical issues and trustworthiness. Beautify faces based on cosmetic understanding using image to image translation.
Tech Stack	AI, Python, GANs
Max Groups	4

Project 3: Cyber Threat Intel (CTI) Chatbot using RAG

Content	
Project Title	Cyber Threat Intel (CTI) Chatbot using RAG
Category	Cyber
Problem Statement	Security analysts drown in unstructured threat reports (PDFs, blogs). They need a system to answer questions like "What are the latest IOCs for APT29?" instantly based on internal documents.
Tech Stack	Python, LangChain, OpenAI GPT-4 or Llama 3 (local), FAISS (Vector DB), Streamlit.
Max Groups	2

Project 4: Deepfake Detection using Vision Transformers

Content	
Project Title	Deepfake Detection using Vision Transformers
Category	Cyber
Problem Statement	CNNs often fail to detect high-quality deepfakes generated by GANs. Vision Transformers (ViT) can capture global pixel dependencies to spot inconsistent artifacts in fake faces.
Tech Stack	Python, PyTorch, Google ViT (Vision Transformer), OpenCV.
Max Groups	3

Project 5: Automated Vulnerability Patching with LLMs

Content	
Project Title	Automated Vulnerability Patching with LLMs
Category	Cyber
Problem Statement	Identifying a bug is only half the battle; fixing it takes time. This project uses LLMs to not only find vulnerabilities (like Buffer Overflows) in C/C++ code but also suggest the corrected code.
Tech Stack	Python, Ollama (Llama 3 or CodeLlama), AST parsing, C/C++ datasets.
Max Groups	3

Project 6: Automated Digital Forensics Reporter

Content	
Project Title	Automated Digital Forensics Reporter
Category	Cyber
Problem Statement	Forensic analysts spend hours writing reports after analyzing disk images. This tool ingests raw forensic logs (e.g., from Autopsy) and generates a human-readable legal case report.
Tech Stack	Python, Autopsy (for data), OpenAI API or Llama 3, RAG (Retrieval Augmented Generation).
Max Groups	3

Project 7: Leveraging GAN-Based Generative Replay to Combat Catastrophic Forgetting in Ma

Content	
Project Title	Leveraging GAN-Based Generative Replay to Combat Catastrophic Forgetting in Malware Classification
Category	Cyber
Problem Statement	Malware detection models are usually trained on fixed datasets and updated when new malware types appear. During such updates, models often forget previously learned malware patterns, a problem known as catastrophic forgetting. Retraining models using all old and new data is not practical due to large data size and storage limitations. This proposes to address this issue by using a GAN-based generative replay approach, where synthetic malware samples are generated to represent older malware classes. These generated samples are combined with new malware data during training to help the model retain past knowledge. The effectiveness of this approach is evaluated by comparing detection accuracy before and after learning new malware classes. To overcome hardware and other resources constraints, the original MalCL framework can be downsized by using a smaller subset of the dataset, fewer malware classes, and reduced training epochs preserving the core generative replay mechanism while ensuring feasible execution on moderate resouces such as laptop.
Tech Stack	Python,PyTorch,Generative Adversarial Networks (GAN),Deep Neural Networks (Classifier),EMBER Malware Dataset (downsized),Anaconda Environment
Max Groups	3

Project 8: CRYSTAL DIFFUSION VARIATIONAL AUTOENCODER FOR PERIODIC MATERIAL

Content	
Project Title	CRYSTAL DIFFUSION VARIATIONAL AUTOENCODER FOR PERIODIC MATERIAL GENERATION
Category	Data
Problem Statement	https://arxiv.org/abs/2110.06197
Tech Stack	Gen AI, Python
Max Groups	4

Project 9: MolGAN: An Implicit Generative Model for Small Molecular Graphs Authors: Nicola De Cao & Thomas Kipf Title: MolGAN: An implicit generative model for small molecular graphs

Content	
Project Title	MolGAN: An Implicit Generative Model for Small Molecular Graphs Authors: Nicola De Cao & Thomas Kipf Title: MolGAN: An implicit generative model for small molecular graphs
Category	Data
Problem Statement	https://arxiv.org/abs/2110.06197
Tech Stack	Gen AI, Python
Max Groups	4

Project 10: Detecting Hallucinations in Large Language Models Using Internal State Semantics

Content	
Project Title	Detecting Hallucinations in Large Language Models Using Internal State Semantics
Category	LLMScan
Problem Statement	Large Language Models (LLMs) are widely used today, but they often give answers that sound confident even when they are wrong. These mistakes, called hallucinations, can be risky when AI is used for learning, healthcare, or decision-making. This project focuses on detecting such hallucinations by looking inside the model, instead of only checking its final answers. By studying the model's internal representations while it generates multiple responses, we can see if the model is consistent or uncertain its understanding really is. This approach helps identify both confused and overconfident wrong answers, making AI systems safer, more reliable, and easier to trust.
Tech Stack	Python + PyTorch + TransformerLens FastAPI + Uvicorn
Max Groups	4

Project 11: LLMSCAN: Causal Scan for LLM Misbehavior Detection

Content	
Project Title	LLMSCAN: Causal Scan for LLM Misbehavior Detection
Category	LLMScan
Problem Statement	Large Language Models (LLMs) sometimes behave incorrectly by producing harmful, biased, or unsafe outputs, even when they appear confident. This paper introduces LLMSCAN, a method that detects such misbehavior by analyzing how different internal components of an LLM cause its final output. Instead of only checking what the model says, LLMSCAN studies how changing or intervening in internal activations affects the model's behavior. By using causal identify hidden sources of misbehavior inside the model, making it easier to audit, debug, and improve the safety of LLMs.
Tech Stack	LLaMA-2 / Mistral+PyTorch+Hugging Face+PyTorch forward hooks+Scaling, zeroing, noise + Pandas+FastAPI+Streamlit
Max Groups	3

Project 12: Psychological Steering in LLMs: An Evaluation of Effectiveness and Trustworthiness

Content	
Project Title	Psychological Steering in LLMs: An Evaluation of Effectiveness and Trustworthiness
Category	LLMScan
Problem Statement	Large Language Models (LLMs) can be influenced by the way prompts are written, especially when they include psychological cues such as authority, emotion, urgency, or social pressure. This paper studies psychological steering, which refers to intentionally guiding LLMs with such cues. The goal is to evaluate how effective these techniques are and whether they affect the trustworthiness of the model's outputs. Understanding psychological steering helps ensure that LLMs remain reliable, unbiased, and safe when interacting with users.
Tech Stack	LLaMA-2 / Mistral+PyTorch+Hugging Face+PyTorch + Transformers Pandas+FastAPI+Streamlit
Max Groups	5

Project 13: AI-Based Dietary Assessment and Nutrient Estimation using multimodal large language models

Content	
Project Title	AI-Based Dietary Assessment and Nutrient Estimation using multimodal large language models
Category	Nutra
Problem Statement	Develop a mobile application that performs dietary assessment from food images and estimates the nutrient content of consumed foods using multimodal large language models.
Tech Stack	Flutter/kotlin,firebase,GPT_Vision_Model,RAG,Redis/celery,JWT
Max Groups	4

Project 14: AI identification of plant diseases and the classification of natural drug compounds

Content	
Project Title	AI identification of plant diseases and the classification of natural drug compounds
Category	Nutra
Problem Statement	Develop a mobile-based artificial intelligence platform for the identification of plant diseases and the classification of natural drug compounds sourced from plants, fungi, and bacteria.
Tech Stack	CNN,ML,Flutter/kotlin,JWT,Redis/celery
Max Groups	3

Project 15: Multimodal platform for NutritionVQA

Content	
Project Title	Multimodal platform for NutritionVQA
Category	Nutra
Problem Statement	Develop a mobile based multimodal platform to capture and reason about the relationships between nutrients, including vitamins, minerals, and macronutrients, and their associated health outcomes
Tech Stack	VLM,Flutter/kotlin,JWT,Redis/celery
Max Groups	4

Project 16: A Deep Learning Web Platform for Sonogram-Based Lactation Stage Classification

Content	
Project Title	A Deep Learning Web Platform for Sonogram-Based Lactation Stage Classification in Dairy Cows
Category	Nutra
Problem Statement	Develop a web-based platform that demonstrates the effectiveness of analysis for dairy production assessment by classifying dairy cows according to lactation status and distinguishing between major productive stages, including the dry period, peak lactation, late lactation, fresh cows, and peri-partum stages
Tech Stack	CNN,ML,MERN,JWT,Redis/celery
Max Groups	3

Project 17: An AI Based Mental Health Chatbot for Stress Detection

Content	
Project Title	An AI Based Mental Health Chatbot for Stress Detection
Category	Neuro
Problem Statement	Stress often goes unnoticed because individuals hesitate to seek professional help or lack access to mental health services. Many people express their emotions indirectly through text messages in daily conversation. There is a need for an automated conversational system that can interpret natural language, detect stress levels, and provide timely support. SilentStress Detector aims to identify stress from user messages using NLP and deep learning and respond with appropriate advice, thus offering early emotional assistance.
Tech Stack	Neuro
Max Groups	2

Project 18: NeuroTalk AI: LLM Based Cognitive & Emotional Pattern Analysis System

Content	
Project Title	NeuroTalk AI: LLM Based Cognitive & Emotional Pattern Analysis System
Category	Neuro
Problem Statement	Mental well being issues often remain unnoticed as individuals may not explicitly express emotional distress or may lack access to professional assessment tools. While people frequently communicate their thoughts and emotions through text, manual interpretation of such data is subjective and inconsistent. Existing mental health assessment systems are either clinically complex, invasive, or unsuitable for large-scale use. There is a need for an automated, ethical, and language-based system that can analyze textual inputs to identify emotional and cognitive patterns indicative of mental well-being without performing medical diagnosis. NeuroTalk AI addresses this need by using LLM-based natural language understanding to provide awareness-driven emotional and cognitive insights from user text.
Tech Stack	Neuro
Max Groups	2

Project 19: SleepSense AI : Sleep, Behavior and Voice Based Mental Well Being Analysis

Content	
Project Title	SleepSense AI : Sleep, Behavior and Voice Based Mental Well Being Analysis
Category	Neuro
Problem Statement	While poor sleep is widely recognized as a risk factor for mental health difficulties, most educational tools fail to combine quantitative sleep metrics, personal subjective experiences, and acoustic speech features in a unified analysis. Clinical research indicates that vocal prosody and acoustic cues correlate with emotional states and mental health conditions yet such analysis is uncommon in undergraduate projects due to perceived complexity. There is a need for a system that integrates sleep data, user-reported textual feedback, and speech acoustics to generate richer insights into mental well-being. SleepSense AI addresses this by combining simplified multimodal techniques within a web environment.
Tech Stack	Neuro
Max Groups	2

Project 20: Learning in Focus: Detecting Behavioral and Collaborative Engagement Using Vision Transformers

Content	
Project Title	Learning in Focus: Detecting Behavioral and Collaborative Engagement Using Vision Transformers
Category	Neuro
Problem Statement	AI-driven approach that leverages Vision Transformers (ViTs) to automatically classify childrens engagement using visual cues such as gaze direction, interaction, and peer collaboration.
Tech Stack	Neuro
Max Groups	2

Project 21: Menta: A Small Language Model for On-Device Mental Health Prediction

Content	
Project Title	Menta: A Small Language Model for On-Device Mental Health Prediction
Category	Neuro
Problem Statement	Menta, the first optimized SLM fine tuned specifically for multi-task mental health prediction from social media data.
Tech Stack	Neuro
Max Groups	1

Project 22: Detecting schizophrenia with 3D structural brain MRI using deep learning (Nature, C

Content	
Project Title	Detecting schizophrenia with 3D structural brain MRI using deep learning (Nature, Challenging project, talks about building a transformer model from scratch)
Category	Neuro
Problem Statement	Detecting schizophrenia with 3D structural brain MRI using deep learning
Tech Stack	Neuro
Max Groups	4

Project 23: Speech-Based Estimation of Schizophrenia Severity

Content	
Project Title	Speech-Based Estimation of Schizophrenia Severity
Category	Neuro
Problem Statement	Speech-based assessment of the schizophrenia spectrum has been widely researched over in the recent past. In this study, we develop a deep learning framework to estimate schizophrenia severity scores from speech using a feature fusion approach that fuses articulatory features with different selfsupervised speech features extracted from pre-trained audio models (Achitecture : Vector Quantized Variational Auto Encoder - VQ-VAE-based articulatory representation learning model)
Tech Stack	Neuro
Max Groups	4

Project 24: LLM-Prop: predicting the properties of crystalline materials using large language models

Content	
Project Title	LLM-Prop: predicting the properties of crystalline materials using large language models
Category	Crystal
Problem Statement	Develop an end to end deep learning system that predicts a crystalline materials band gap (in electron volts, eV) from its natural-language crystal structure description. These descriptions can be generated from a CIF (Crystallographic Information File) using Robocrystallographer, a tool that automatically produces a deterministic, human readable text description of a crystal structure given its CIF file. Using an instructor-provided dataset (or a subset of TextEdge from the LLM Prop work, where LLM = Large Language Model), create training/validation/test splits, preprocess the text, and fine-tune a pretrained Transformer encoder (recommended: the encoder of T5 small, where T5 = Text to Text Transfer Transformer) with a small regression head to output the band gap value. Evaluate and report performance using MAE (Mean Absolute Error) on a held-out test set, and provide a simple demo such as a CLI (Command Line Interface) program where a user pastes a crystal description and receives the predicted band gap. Submit a runnable code repository, a short report describing the dataset, model, training setup, and results, and the demo script; benchmarking against simpler baselines and ablation/preprocessing studies are optional if time permits.
Tech Stack	Gen AI, Mern Stack, Python
Max Groups	3

Project 25: Quokka: An Open-source Large Language Model ChatBot for Material Science

Content	
Project Title	Quokka: An Open-source Large Language Model ChatBot for Material Science
Category	Crystal
Problem Statement	Develop a specialized chatbot for materials science, leveraging the Llama family (or other) language model, and continuing pre-training on the expansive research articles in the materials science domain from the S2ORC (or other relevant) dataset. The methodology involves an initial pretraining phase on domain-specific papers, followed by an instruction-tuning process to refine the chatbot's capabilities. The chatbot is designed to assist researchers, educators, and students by providing instant, context-aware responses to queries in the field of materials science.
Tech Stack	Gen AI, Mern Stack, Python
Max Groups	3

Project 26: CrysText: A Generative AI Approach for Text-Conditioned Crystal Structure Generation

Content	
Project Title	CrysText: A Generative AI Approach for Text-Conditioned Crystal Structure Generation using LLM
Category	Crystal
Problem Statement	Generating crystal structures directly from textual descriptions marks a pivotal advancement in materials informatics, offering a streamlined pathway from concept to discovery. Integrating generative models into Crystal Structure Prediction (CSP) presents a transformative opportunity to enhance efficiency and innovation. While large language models (LLMs) excel at understanding and generating text, their potential in materials discovery remains largely unexplored. Here, we introduce CrysText, an advanced approach for generating crystal structures from simple text prompts, conditioned on material composition and space group number. Leveraging Llama-3.1-8B fine-tuned with Quantized Low-Rank Adaptation (QLoRA), our approach enables the efficient and scalable generation of CIF-formatted structures directly from input descriptions, eliminating the need for post-processing and ensuring effective fine-tuning with rapid inference. Evaluations on the MP-20 benchmark dataset demonstrate high structure match rates and effective RMSE metrics, showcasing the framework's ability to generate crystal structures that faithfully adhere to specified compositions and crystal symmetries. By conditioning on energy above the hull, we further demonstrate the potential of CrysText to generate stable crystal structures. Our work highlights the transformative role of LLMs in text-prompted inverse design, accelerating the discovery of new materials.
Tech Stack	Gen AI, Mern Stack, Python
Max Groups	3

Project 27: Automated Seizure Detection using Transformer Models on Multi-Channel EEGs

Content	
Project Title	Automated Seizure Detection using Transformer Models on Multi-Channel EEGs
Category	Cyber
Problem Statement	Epilepsy affects 50 million people worldwide, with recurrent seizures requiring early prediction to prevent severe complications. Scalp EEGs enable noninvasive seizure forecasting, but manual annotation is labor-intensive, costly, and inconsistent across clinicians. Automated deep learning solutions are essential for efficient multi-channel EEG analysis. This hybrid CNN-Transformer framework uses CNN for local temporal-spatial feature extraction from raw EEG and Transformer for long-range dependencies. It enables timely seizure prediction, outperforming SOTA baselines
Tech Stack	CNN (EEGNet), Transformer Encoder, Frameworks: PyTorch, MNE-Python, Conda Environment
Max Groups	3

Project 28: Deep learning models for the prediction of acute postoperative pain in PACU

Content	
Project Title	Deep learning models for the prediction of acute postoperative pain in PACU
Category	Cyber
Problem Statement	Acute postoperative pain affects 30-50% of patients in PACU, prolonging recovery and increasing opioid use, with current prediction methods relying on subjective clinical judgment without integrating real-time intraoperative vital signs. DoseFormerýýa novel Graph Transformer Attention Network (GAT)ýýjointly processes static patient features (demographics, medical history) and dynamic time-series vitals across surgical phases via multi-head attention mechanisms. This transformer-based deep learning approach, demonstrated through code implementation, enables automated perioperative risk stratification and lays foundation for neurosurgery extension
Tech Stack	Transformers, Graph Neural Networks, Time-Series Analysis, Multi-Modal Fusion, Attention Mechanisms and PyTorch
Max Groups	3

Project 29: ABFormer: A Transformer-based Model to Enhance Antibody-Drug Conjugates Activity Prediction through Contextualized Antibody-Antigen Embedding

Content	
Project Title	ABFormer: A Transformer-based Model to Enhance Antibody-Drug Conjugates Activity Prediction through Contextualized Antibody-Antigen Embedding
Category	Drug Discovery
Problem Statement	These research papers have been developed by students of DrugParadigm (K-Hub), and several of them are currently under peer review. To protect the originality, intellectual property, and publication priority of this work, please ensure that no content , including abstracts, figures, results, or methodological details is shared on public platforms, repositories, social media, or preprint servers. Internal use for learning, discussion, and feedback is okay and the use of AI tools such as ChatGPT or similar assistants is fine as well. Any external sharing must receive prior written approval from DrugParadigm until the peer-review and publication process is complete.
Tech Stack	GNNs , Transformers , Diffusion , GVP , GCN , RDkit , Auto Dock , SE3 transformers
Max Groups	4

Project 30: Macro Equi Diff (MED): De Novo Macrocycles Generation Using Equivariant Diffusion

Content	
Project Title	Macro Equi Diff (MED): De Novo Macrocycles Generation Using Equivariant Diffusion
Category	Drug Discovery
Problem Statement	These research papers have been developed by students of DrugParadigm (K-Hub), and several of them are currently under peer review. To protect the originality, intellectual property, and publication priority of this work, please ensure that no content , including abstracts, figures, results, or methodological details is shared on public platforms, repositories, social media, or preprint servers. Internal use for learning, discussion, and feedback is okay and the use of AI tools such as ChatGPT or similar assistants is fine as well. Any external sharing must receive prior written approval from DrugParadigm until the peer-review and publication process is complete.
Tech Stack	GNNs , Transformers , Diffusion , GVP , GCN , RDkit , Auto Dock , SE3 transformers
Max Groups	4

Project 31: CrossPPI: A Cross Fusion Based Model for Protein Protein Binding Affinity Prediction

Content	
Project Title	CrossPPI: A Cross Fusion Based Model for Protein Protein Binding Affinity Prediction
Category	Drug Discovery
Problem Statement	These research papers have been developed by students of DrugParadigm (K-Hub), and several of them are currently under peer review. To protect the originality, intellectual property, and publication priority of this work, please ensure that no content , including abstracts, figures, results, or methodological details is shared on public platforms, repositories, social media, or preprint servers. Internal use for learning, discussion, and feedback is okay and the use of AI tools such as ChatGPT or similar assistants is fine as well. Any external sharing must receive prior written approval from DrugParadigm until the peer-review and publication process is complete.
Tech Stack	GNNs , Transformers , Diffusion , GVP , GCN , RDkit , Auto Dock , SE3 transformers
Max Groups	4

Project 32: SE(3) PROTACs: Geometric Deep Learning for PROTAC Degradation Prediction

Content	
Project Title	SE(3) PROTACs: Geometric Deep Learning for PROTAC Degradation Prediction
Category	Drug Discovery
Problem Statement	These research papers have been developed by students of DrugParadigm (K-Hub), and several of them are currently under peer review. To protect the originality, intellectual property, and publication priority of this work, please ensure that no content , including abstracts, figures, results, or methodological details is shared on public platforms, repositories, social media, or preprint servers. Internal use for learning, discussion, and feedback is okay and the use of AI tools such as ChatGPT or similar assistants is fine as well. Any external sharing must receive prior written approval from DrugParadigm until the peer-review and publication process is complete.
Tech Stack	GNNs , Transformers , Diffusion , GVP , GCN , RDkit , Auto Dock , SE3 transformers
Max Groups	4